Figure 26-1 illustrates a systematic way of analyzing this situation. Suppose the device is tested on two groups: several hundred volunteers known to be healthy (nontarget), and several hundred volunteers known to have cancer (target). Figures (a) & (b) show these test results displayed as histograms. The healthy subjects generally produce a lower number than those that have cancer (good), but there is some overlap between the two distributions (bad).

As discussed in Chapter 2, the histogram can be used as an estimate of the **probability distribution function (pdf)**, as shown in (c). For instance, imagine that the device is used on a randomly chosen healthy subject. From (c), there is about an 8% chance that the test result will be 3, about a 1% chance that it will be 18, etc. (This example does not specify if the output is a real number, requiring a *pdf*, or an integer, requiring a *pmf*. Don't worry about it here; it isn't important).

Now, think about what happens when the device is used on a patient of unknown health. For example, if a person we have never seen before receives a value of 15, what can we conclude? Do they have cancer or not? We know that the probability of a healthy person generating a 15 is 2.1%. Likewise, there is a 0.7% chance that a person with cancer will produce a 15. If no other information is available, we would conclude that the subject is three times as likely not to have cancer, as to have cancer. That is, the test result of 15 implies a 25% probability that the subject is from the target group. This method can be generalized to form the curve in (d), the probability of the subject having cancer based only on the number produced by the device [mathematically, $pdf_t / (pdf_t + pdf_{nt})$].

If we stopped the analysis at this point, we would be making one of the most common (and serious) errors in target detection. Another source of information must usually be taken into account to make the curve in (d) meaningful. This is the relative number of targets versus nontargets in the population to be tested. For instance, we may find that only one in one-thousand people have the cancer we are trying to detect. To include this in the analysis, the amplitude of the nontarget pdf in (c) is adjusted so that the area under the curve is 0.999. Likewise, the amplitude of the target pdf is adjusted to make the area under the curve be 0.001. Figure (d) is then calculated as before to give the probability that a patient has cancer.

Neglecting this information is a serious error because it greatly affects how the test results are interpreted. In other words, the curve in figure (d) is drastically altered when the prevalence information is included. For instance, if the fraction of the population having cancer is 0.001, a test result of 15 corresponds to only a 0.025% probability that this patient has cancer. This is very different from the 25% probability found by relying on the output of the machine alone.

This method of converting the output value into a probability can be useful for understanding the problem, but it is not the main way that target detection is accomplished. Most applications require a yes/no decision on